



LECTURE 6

Sprint Definition, Sprint Planning

COEN / ELEC 390

Acknowledgement

- The concepts in these slides are based on the textbook:

Essential Scrum, A Practical Guide to the Most Popular Agile Process
Kenneth S. Rubin, Addison-Wesley, 2013.

Sprints

- Sprints are timeboxed iterations with specific start and end dates; each sprint should lead to a potentially shippable release.
- The team should include in a sprint on only those items that it believes it can start and finish within the sprint.
- Timeboxed iterations prioritize essential work, demonstrate progress through validation, prevent unnecessary perfectionism, and motivate closure with deadlines.

Sprint Duration

- A sprint duration should be short (1 to 4 weeks) to ease of planning, promote fast feedback, and allow for frequent checkpoints
- Consistent duration:
 - The sprint duration should not change unless there is a compelling reason (e.g., moving from four-week sprints to two-week sprints, annual holidays, etc.)
 - Consistent duration helps estimate the team velocity (work/sprint), reduces overhead, allows synchronization across multiple teams, eases release planning how many sprints in the future).

No Goal-altering Changes

- A sprint must have a clear, single-focus, such as:
 - Support initial report generation
 - Load and curate North America map data
 - Demonstrate the ability to send a text message through an integrated software
- The team commits to meeting the goal by the end of the sprint, and the product owner commits to not altering the goal during the sprint.
- Changes can affect the project and the motivation of the team.

Definition of Done

The Definition of Done is a clear and shared understanding among the team of what it means for a product backlog item to be considered complete.

TABLE 4.1 Example Definition-of-Done Checklist

Definition of Done	
☐	Design reviewed
☐	Code completed
☐	Code refactored
☐	Code in standard format
☐	Code is commented
☐	Code checked in
☐	Code inspected
☐	End-user documentation updated
☐	Tested
☐	Unit tested
☐	Integration tested
☐	Regression tested
☐	Platform tested
☐	Language tested
☐	Zero known defects
☐	Acceptance tested
☐	Live on production servers

Source: Essential Scrum, A Practical Guide to the Most Popular Agile Process by Kenneth S. Rubin, Addison-Wesley, 2013.

Sprint Planning

- Sprint planning is a recurring activity that takes place at the beginning of each sprint.
- The full Scrum team collaborates during sprint planning.
 - The product owner shares the initial sprint goal, presents the prioritized product backlog, and answers any questions.
 - The development team works diligently to determine what it can deliver and then makes a realistic commitment at the end of sprint planning.
 - The ScrumMaster, acting as the Scrum team coach, observes the planning activity, asks probing questions, and facilitates to ensure a successful result.

Sprint Planning Principles

- Do not assume you can get things right up front
 - Produce some of the planning artifacts early on to achieve a good balance between up-front and just-in-time planning.
- Up-front planning should be helpful not excessive
- Keep planning options open until the last responsible moment
- Focus more on adapting & re-planning than conforming to a plan

Planning Process

- Sprint planning participants should have the chance to review and discuss potential value-generating alternatives, deciding what is feasible based on the team's capacity, expected velocity, and any known constraints.
- To build confidence in what can be achieved, the development team will create a plan to reach the sprint goal.
- This plan involves breaking down the sprint backlog items into tasks, each sized by effort hours, with the ideal maximum for any task being 8 hours.

Selecting Product Backlog Items

- The team needs to start with the highest priority item and then proceed to the next one, continuing in this manner.
- If the team cannot commit to the next highest-priority item, it will choose the next suitable item that seems feasible within the given constraints.
- The "start only what you can finish" rule limits work in progress (WIP) and recognizes that starting tasks without completing them leads to various forms of waste.

Example of a sprint backlog

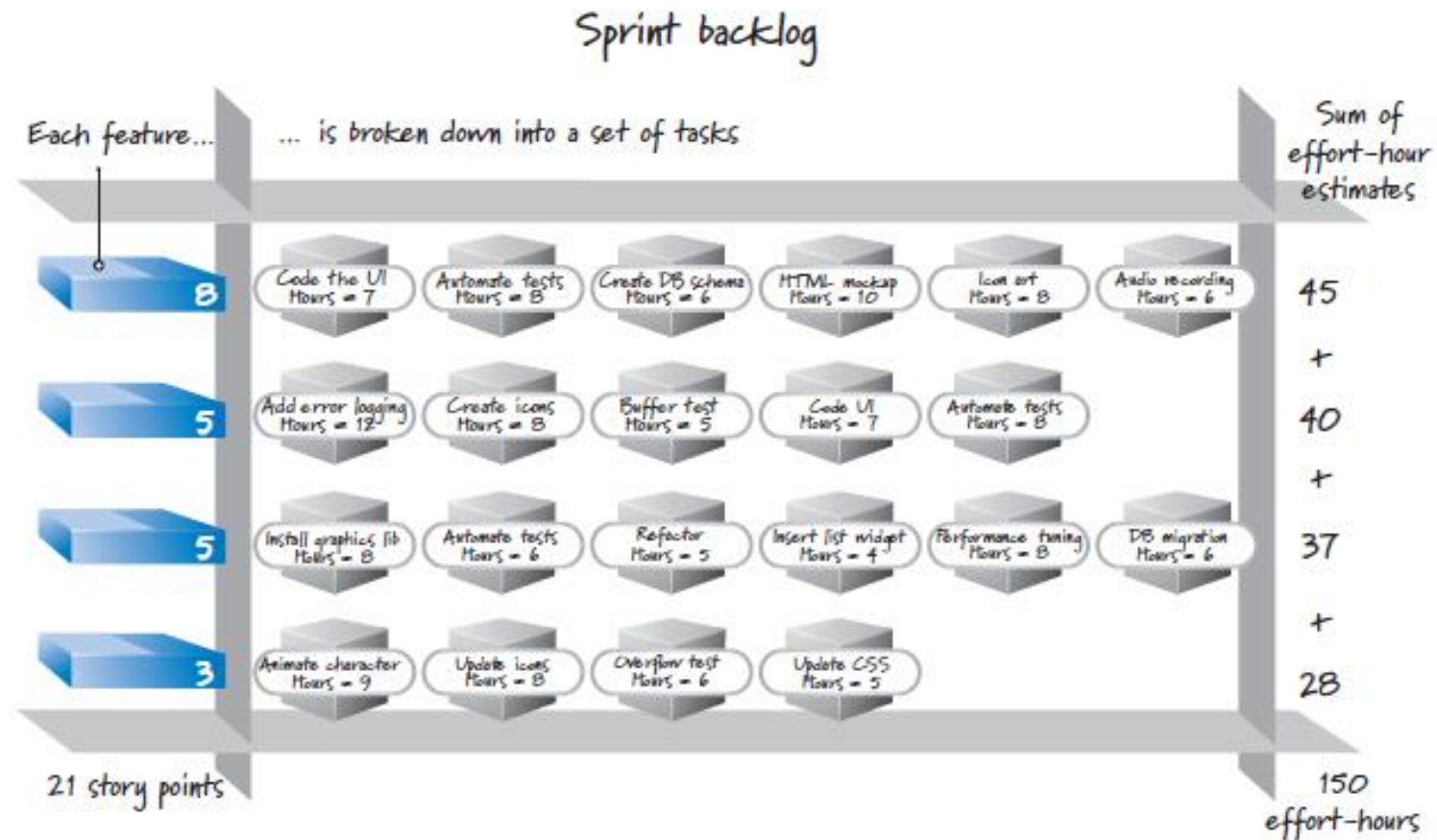


FIGURE 19.6 Sprint backlog showing PBIs and task plan

The Concept of Velocity?

- Velocity is the amount of work completed in each sprint.
- Velocity measures output (the size of what was delivered), not outcome (the value of what was delivered).
- Velocity is used for two important purposes:
 - Essential concept for Scrum planning (number of sprints/feature, number of person-hours/task and hence, number of tasks in a sprint, etc.)
 - A diagnostic metric that the team can use to evaluate and improve the team and the process

Overview

Velocity is measured as work per unit of time.

Here the unit of time is a sprint.

Work may be measured in ideal days or story points.

- Duration of the project = size/velocity

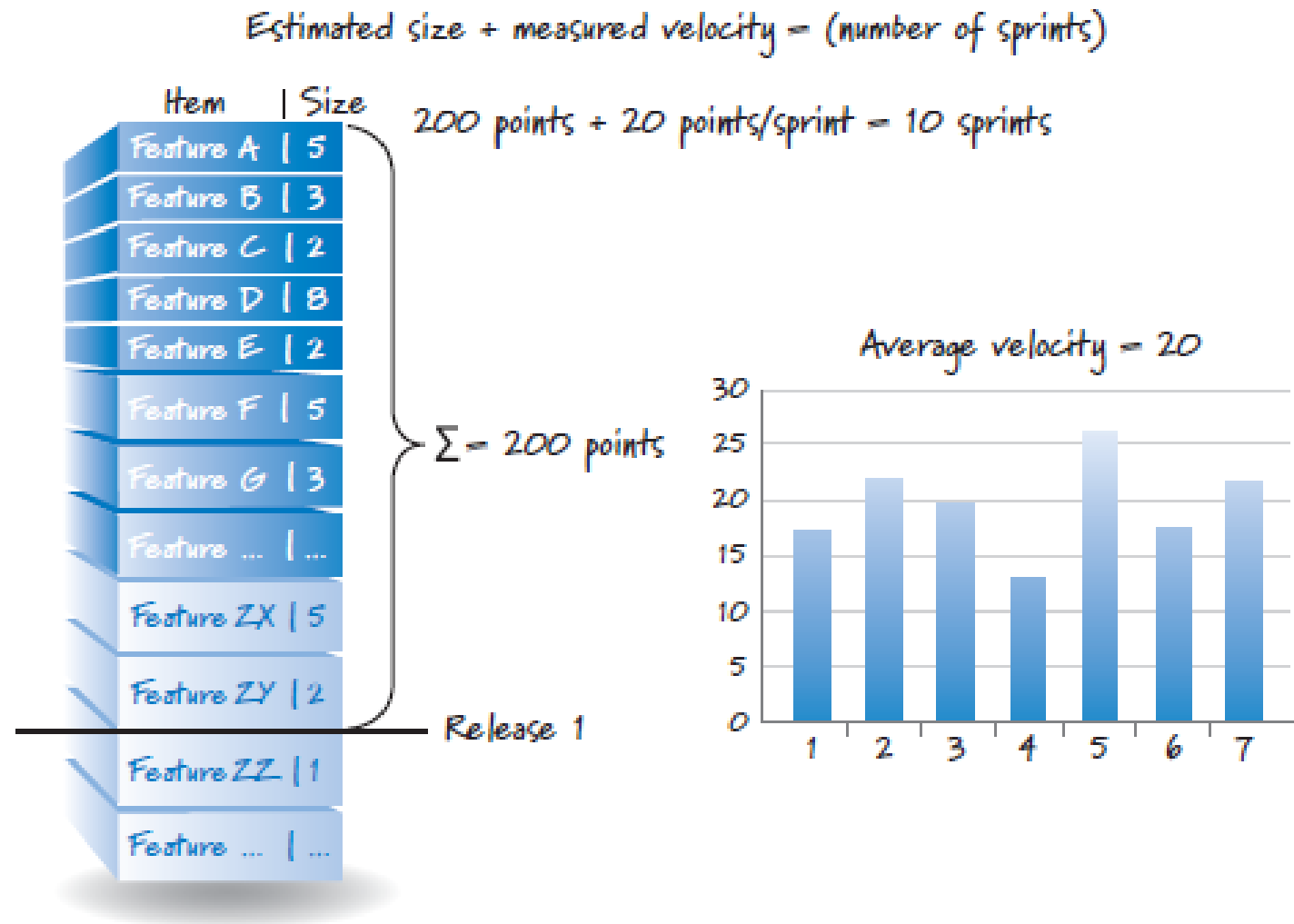


FIGURE 7.1 The relationship among size, velocity, and duration

Affecting Velocity

- If the team continuously inspects and adapts (constantly improving), its velocity should steadily increase.
- Implementing new tools and training can positively impact velocity although these changes often lead to a temporary dip as the team adjusts.
- Additionally, while working overtime may yield short-term gains, the long-term repercussions usually far outweigh these benefits.
- Velocity is used as a planning tool and as a team diagnostic metric. It should not be used as a performance metric in an attempt to judge team productivity. Why?

PBI Estimation Units

- Story Points:
 - A story point is a unit of measure to estimate the relative effort required to complete a user story or task.
 - Instead of measuring time, story points take into account factors like complexity, risk, and the amount of work involved.
 - Story points represent effort associated with the item from the development team's perspective.
- Ideal Days:
 - Represent the number of effort-days or person-days needed to complete a story.
 - An ideal day is when we spend all our time working on the project.
 - Ideal days are not calendar days.
 - Use ideal hours to estimate tasks.

Product Backlog Items Estimates

- Most teams choose to assign numeric size estimates to backlog items, using either story points or ideal days.
- Some measure the number of completed PBIs in a sprint rather than the total size of the completed items.
- However, many still prefer to estimate the size of PBIs for several reasons:
 - Not all PBIs will be of equal size.
 - It can take time for teams to break down PBIs so that they are similarly sized.
 - Teams may need to split stories at awkward points to achieve equal sizing.
 - The estimation process itself is a valuable activity for the team.

PBI Estimation Principles

- **Estimate as a team:** The people who will do the work collectively provide the estimates.
- **Estimates are not commitments:** The estimates are a realistic measure of how big something is. We don't want them artificially inflated due to external influences.
- **Estimates should be accurate without being overly precise.** Or else, too much effort put into inflated precision.
- **Estimate PBIs using relative sizes,** not absolute sizes. People are excellent at relative (not absolute) size estimation.

Planning Poker

- Planning Poker is a technique used to estimate PBIs
- The cards contains numbers that the team uses to estimate the size of a BPI
- The frequently used scale is based in part on a modified Fibonacci sequence: 1, 2, 3, 5, 8, 13, 20, 40, and 100, as well as infinity, ?, and Pi.
- Small items are assigned small numbers 1 and 2, etc. 100 is assigned to large epics.
- Infinity represents a very large item.
- Pi does not mean 3.14, but “I am hungry and I want to get some pie!”

Sizing PBI Using Planning Poker

1. The product owner **selects** a PBI to be estimated and reads the item to the team.
2. Dev team members **discuss** the item and ask clarifying questions to the product owner, who answers the questions.
3. Each estimator **privately** selects a card representing their estimate.
4. Once each estimator has made a private selection, all private estimates are simultaneously **exposed** to all estimators.
5. If everyone selects the **same** card, there is consensus, and that consensus number becomes the PBI estimate.
6. If the estimates are not the same, the team members engage in a focused **discussion** to expose assumptions and misunderstandings.
7. After the discussion, return to **step 3** and repeat until consensus is reached.